

ST LOUIS LAMBERT, MO, USA

WMO: 724340

Lat: 38.753N

Lon: 90.374W

Elev: 531

StdP: 14.42

Time Zone: -6.00 (NAC)

Period: 94-19

WBAN: 13994

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	6.8	12.2	-4.8	4.3	9.4	0.1	5.6	14.3	26.1	31.4	24.0	33.6	9.4	300	0.507

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	16.8	96.1	76.8	93.5	76.2	91.2	75.1	79.5	91.0	78.2	89.5	77.0	87.9	9.9	250

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
76.3	139.9	86.1	74.9	133.3	85.2	73.5	127.1	83.8	43.3	90.9	42.0	89.5	40.7	87.8	85.1

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
23.2	19.9	18.0	1.2	100.4	5.7	3.2	-2.9	102.7	-6.2	104.6	-9.4	106.4	-13.6	108.7		
			0.0	81.6	5.5	1.7	-4.0	82.8	-7.2	83.7	-10.3	84.6	-14.3	85.9		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	57.8	32.3	36.7	46.8	57.9	67.7	76.6	80.3	79.0	71.4	59.5	46.9	36.7
	DBStd	19.13	11.89	11.64	11.24	9.61	8.20	6.35	5.76	5.44	7.53	9.30	10.03	10.51
	HDD50	1804	559	393	195	33	1	0	0	0	0	21	174	428
	HDD65	4379	1014	792	571	253	71	5	0	0	27	222	545	879
	CDD50	4639	10	22	97	270	550	798	940	899	641	315	82	15
	CDD65	1736	0	0	8	40	154	352	475	434	218	52	3	0
	CDH74	17735	0	4	65	352	1303	3586	5477	4519	1999	407	23	0
	CDH80	7381	0	0	7	83	400	1493	2593	1966	738	99	2	0
Wind	WSAvg	8.8	9.8	9.9	10.2	10.4	8.9	8.0	7.4	7.1	7.0	8.4	9.1	9.3
Precipitation	PrecAvg	41.6	2.4	2.2	3.1	4.6	5.0	4.4	3.8	3.4	3.3	3.2	3.5	2.6
	PrecMax	58.1	8.4	5.0	6.9	11.0	11.1	11.5	8.0	6.1	10.2	13.1	7.4	11.1
	PrecMin	29.1	0.7	0.4	1.1	1.9	1.0	0.6	0.7	0.9	0.4	0.6	0.7	0.4
	PrecStd	7.9	1.7	1.2	1.3	2.2	2.6	2.6	1.9	1.4	2.1	2.4	1.9	2.1
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	67.8	73.0	80.4	86.8	91.1	96.1	101.7	99.2	94.2	87.5	77.4	68.4
		MCWB	58.1	59.2	62.5	67.1	71.9	74.9	75.7	76.7	74.5	70.9	65.1	60.5
	2%	DB	60.7	66.4	75.0	82.2	87.5	93.3	97.0	95.5	90.9	82.7	71.3	62.3
		MCWB	53.7	55.8	60.5	64.8	71.1	75.5	77.4	76.6	73.5	67.8	59.6	56.1
	5%	DB	55.3	60.8	69.9	78.1	84.7	90.9	94.0	92.5	87.7	78.7	67.0	57.2
		MCWB	48.9	52.1	58.0	63.0	69.8	74.6	76.9	76.0	72.0	65.7	57.9	50.7
	10%	DB	49.5	55.2	64.8	73.9	81.5	88.3	91.3	89.4	84.2	74.5	63.0	52.2
		MCWB	43.4	48.2	55.1	61.1	68.0	73.4	76.0	74.5	70.2	63.6	55.5	46.4
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	60.4	62.0	65.9	70.1	76.1	79.3	81.3	81.3	77.5	73.2	66.3	62.3
		MCDB	65.8	68.8	76.8	82.1	86.0	89.9	93.1	92.6	88.7	83.6	74.6	67.6
	2%	WB	55.8	58.1	62.5	67.4	73.4	77.7	79.7	79.5	75.5	70.8	62.5	57.3
		MCDB	59.4	64.5	71.5	77.2	83.5	89.3	91.9	90.3	85.9	79.2	67.9	61.3
	5%	WB	49.7	53.2	59.8	65.2	71.7	76.3	78.4	77.9	74.3	68.3	59.7	51.9
		MCDB	55.1	59.1	68.5	74.7	81.3	87.5	90.2	88.4	83.9	75.8	65.7	55.4
	10%	WB	43.6	47.6	56.2	62.9	69.9	74.9	77.4	76.4	72.6	65.2	56.0	46.7
		MCDB	48.9	55.2	63.8	72.3	79.1	85.4	88.7	86.5	81.7	73.0	62.5	51.1
Mean Daily Temperature Range	5% DB	MDBR	15.4	16.6	18.4	19.3	17.6	17.0	16.8	17.1	18.5	18.4	16.8	14.7
		MCDBR	23.9	26.5	24.6	24.4	20.2	19.3	19.0	19.9	20.5	22.0	22.6	21.6
		MCWBR	17.7	18.2	14.1	12.9	9.3	7.4	6.7	6.9	7.9	10.7	14.7	15.8
	5% WB	MCDBR	22.5	24.2	21.5	21.0	18.1	17.6	17.4	17.8	17.7	18.7	19.8	19.8
		MCWBR	18.4	18.3	13.9	12.9	9.5	7.9	7.5	7.6	7.8	10.8	15.1	16.7
Clear-Sky Solar Irradiance	taub	0.303	0.314	0.346	0.401	0.439	0.447	0.475	0.464	0.409	0.350	0.309	0.301	
	taud	2.474	2.437	2.412	2.253	2.210	2.222	2.143	2.173	2.301	2.473	2.530	2.497	
	Ebn at Noon	278	290	290	278	268	266	257	256	265	273	274	271	
	Edh at Noon	25	30	34	43	45	45	48	45	38	29	24	23	
All-Sky Solar Radiation	RadAvg	647	875	1185	1567	1761	2009	1971	1761	1507	1080	731	545	
	RadStd	72	88	112	92	153	154	136	102	95	119	75	59	

Historical Trends

		DBAvg	Heating			Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP		HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	
(46)	Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	
(47)	Regional (31 neighbors)	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	

Nomenclature: See separate page